

Stochastic Processes II:

Brownian Motion & Geometric Brownian Motion

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I. Brownian Motion

1.1 Random Walks and Brownian Motion

(!) Remember: one could provide a RW proper dimensions of time and space, i.e. $t = n\Delta t$ and $\tilde{X} = X\Delta x$.

Consider the symmetric simple RW with $p = q = 1/2$ and the scale transformed variables of time and space above. Using that $\Delta x^2 \equiv \Gamma \Delta t$ (as shown in W6), one directly finds that ($n \rightarrow t, n+1 \rightarrow t + \Delta t$)

$$\tilde{X}_{t+\Delta t} - \tilde{X}_t = \sqrt{\Gamma \Delta t} Z_t, \quad Z_t = \begin{cases} +1 & \text{with prob. } \frac{1}{2} \\ -1 & \text{with prob. } \frac{1}{2}. \end{cases} \quad (1)$$

As a consequence,

- Average displacements,

$$\mathbb{E}[\tilde{X}_{t+\Delta t} - \tilde{X}_t] = 0 \quad (2)$$

- Variance of the displacements,

$$\mathbb{V}[X_{t+\Delta t} - X_t] \equiv \mathbb{E}[(X_{t+\Delta t} - X_t)^2] - \mathbb{E}[X_{t+\Delta t} - X_t]^2 = \Gamma \Delta t \quad (3)$$

1.1 Random Walks and Brownian Motion

It directly follows that ($\Delta t = t/n$)

$$\tilde{X}_t - \tilde{X}_0 = \sqrt{\frac{\Gamma t}{n}} \sum_{\ell=0}^{n-1} Z_\ell \equiv W_t^{(n)}. \quad (4)$$

The **Wiener process** W_t (time-continuous) is defined as the continuous limit (for fixed t , $\Delta t \rightarrow 0$ or $n \rightarrow \infty$),

$$W_t = \lim_{n \rightarrow \infty} W_t^{(n)} = \lim_{n \rightarrow \infty} \sqrt{\frac{\Gamma t}{n}} \sum_{\ell=0}^{n-1} Z_\ell \sim \text{“Brownian motion”}. \quad (5)$$

W_t has the statistical properties of a RW in continuous time, i.e.,

$$\mathbb{E}[W_t^{(n)}] = 0 \Rightarrow \mathbb{E}[W_t] = 0 \quad (6)$$

$$\mathbb{V}[W_t^{(n)}] = \Gamma t \Rightarrow \mathbb{V}[W_t] = \Gamma t. \quad (7)$$

Note

It is common to take $\Gamma = 1$, so that $\mathbb{E}[W_t] = 0$ and $\mathbb{V}[W_t] = t$.

1.1 Random Walks and Brownian Motion

Recall – The central limit theorem (CLT)

Given the set of independent and identically distributed (i.i.d) random variables $X_1, X_2, \dots, X_n$ with $\mathbb{E}[X_i] = \mu$, $\mathbb{V}[X_i] = \sigma^2$ for all $i = 1, \dots, n$, then

$$S_n \equiv \sum_{i=1}^n X_i \stackrel{n \gg 1}{\approx} \mathcal{N}(n\mu, n\sigma^2). \quad (8)$$

The consequence

By definition, then,

$$W_t^{(n)} = \sqrt{\frac{t}{n}} \sum_{\ell=0}^{n-1} Z_\ell \stackrel{n \gg 1}{\approx} \mathcal{N}(0, t). \quad (9)$$

This obviously means that the continuous approximation of a symmetric random walk (diffusion), is described through the normal distribution.

1.2 Introduction to SDEs & Brownian Motion

The Wiener process might be written also in differential form,

$$\lim_{\Delta t \rightarrow 0} (\tilde{X}_{t+\Delta t} - \tilde{X}_t) = \lim_{\Delta t \rightarrow 0} \sqrt{\Gamma \Delta t} Z_t \iff \boxed{dX_t = dW_t}, \quad (10)$$

where

$$\mathbb{E}[dW_t] = 0, \quad \mathbb{V}[dW_t] = dt. \quad (11)$$

The Wiener process then is

$$W_t = \int_0^t dW_s \quad (\text{It\^o integral, stochastic integral}) \quad (12)$$

Stochastic differential equation (SDE)

A stochastic differential equation (SDE) is a differential equation in which at least one of the terms is a stochastic process, meaning its solution is also a stochastic process. For example, any differential equation in which the Wiener process appears, is a SDE. These are usually written as

$$dX_t = f(t, X_t)dt + g(t, X_t)dW_t. \quad (13)$$

1.2 Introduction to SDEs & Brownian Motion

The presence of the Wiener process in SDEs alter the standard rules of calculus used to find solutions to usual ordinary differential equations (ODEs). This is because simply because of the presence of the Wiener process, for which $dW_t^2 \sim dt$.

- We will not go into the details of this (it requires a decent amount of time)
- The Wiener process is not differentiable anywhere.
- The Wiener process introduces differences of order $\sqrt{\Delta t} \gg \Delta t$ (they need to be accounted for)!

Brownian motion with drift

A very common example is the Brownian motion with drift,

$$dX_t = \mu dt + \sigma dW_t. \quad (14)$$

One may show that $\mathbb{E}[X_t] = \mu t$ ($X_{t=0} = 0$) and $\mathbb{V}[X_t] = \sigma^2 t$, meaning $X_t \sim \mathcal{N}(\mu t, \sigma^2 t)$.

II. Geometric Brownian Motion

2.1 Introduction to GBM

Geometric brownian motion (GBM) is the simplest model for asset prices, S_t , one may come up with;

- Asset prices remain positive or get absorbed into null value (valueless).
- Fluctuations are stronger the bigger the price (high S_t means strong fluctuations, low S_t means weak fluctuations).

It is modelled through the SDE,

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_{t=0} = S_0. \quad (15)$$

GMB has expected value and variance

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \quad \mathbb{V}[S_t] = \mathbb{E}[S_t^2] - \mathbb{E}[S_t]^2 = S_0^2 (e^{\sigma^2 t} - 1) e^{2\mu t}. \quad (16)$$

A question for the audience

Would you put your hand on the fire and say

$S_t \sim \mathcal{N}(S_0 e^{\mu t}, S_0^2 (e^{\sigma^2 t} - 1) e^{2\mu t})$? **If your answer is yes, you will burn!**

2.1 Introduction to GBM

Indeed, using Itô calculus, one may show that

$$d(\log S_t) = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t, \quad (\text{log-normality again!}) \quad (17)$$

being thus the formal solution to GBM,

$$\boxed{S_t = S_0 e^{(\mu - \sigma^2/2)t} e^{\sigma W_t}} \Rightarrow \log \left(\frac{S_t}{S_0} \right) \sim \mathcal{N} \left(\left(\mu - \frac{\sigma^2}{2} \right) t, \sigma^2 t \right), \quad (18)$$

meaning asset prices are distributed log-normally (logarithmic returns are normally distributed)!

Standardization

A direct consequence is that, at any time t ,

$$X_t \equiv \frac{\log(S_t/S_0) - \left(\mu - \frac{\sigma^2}{2} \right) t}{\sigma \sqrt{t}} \sim \mathcal{N}(0, 1). \quad (19)$$

III. Implementing GBM & testing with real data

3.1 Implementing and testing of GBM

Solving SDEs (numerically) involves a discretization scheme. The simplest one may think of is the Euler-Maruyama scheme.

Euler-Maruyama scheme

The SDE $dX_t = f(t, X_t)dt + g(t, X_t)dW_t$ can be approximated over $t \in [0, T]$ through the Markov chain

$$X_{n+1} = X_n + f(t_n, X_n)\Delta t + g(t_n, X_n)\Delta W_n, \quad n = 0, \dots, N-1 \quad (20)$$

where $\Delta t = T/N$ and $\Delta W_n \sim \mathcal{N}(0, \Delta t)$ i.i.d (ideally, N large).

Euler-Maruyama scheme of GBM

The solution to GBM is then approximated by the Markov chain,

$$S_{n+1} = (1 + \mu\Delta t + \sigma\Delta W_n)S_n, \quad (21)$$

with $\Delta W_n \sim (0, \Delta t)$ i.i.d.

3.1 Implementing and testing of GBM

Access the GitHub repo [here](#).

